

28. Boundary Detection using Convolution Kernel

Aim

To find the boundary of an image using a convolution kernel in OpenCV.

Procedure

1. Read the input image.
2. Convert the image to grayscale.
3. Define a boundary convolution kernel.
4. Apply the kernel using filter2D().
5. Display the original and boundary images.

Program

```
import cv2
import numpy as np

img = cv2.imread("image.jpg")

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

kernel = np.array([
    [-1, -1, -1],
    [-1, 8, -1],
    [-1, -1, -1]
])

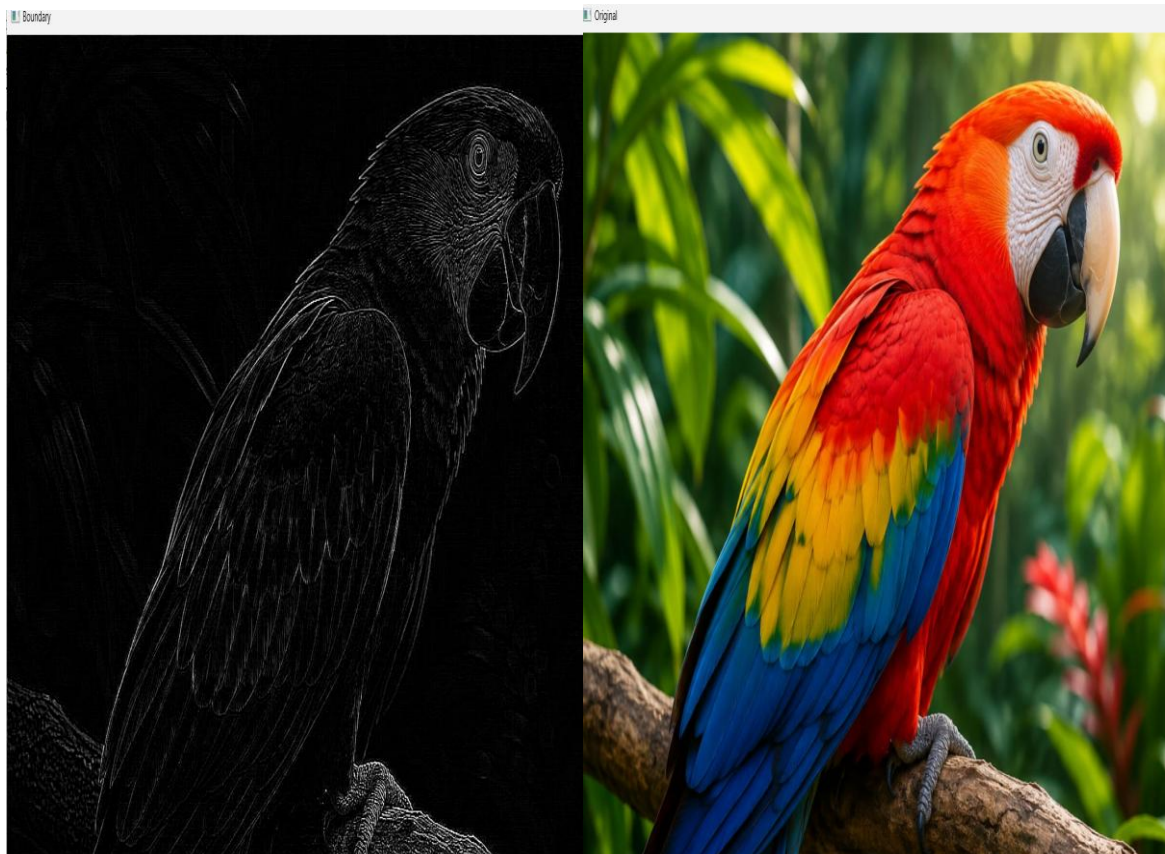
boundary = cv2.filter2D(gray, -1, kernel)

cv2.imshow("Original", img)
cv2.imshow("Boundary", boundary)
```

```
cv2.waitKey(0)
```

```
cv2.destroyAllWindows()
```

output



Result

The boundary of the given image was successfully detected using a convolution kernel.